SET 2: metal-nonmetal compound; three or more elements

Rules: 1. the metallic element is named first
2. the chemical radical is named second

examples $FeSO_4$ = iron sulfate
$Mg(NO_3)_2$ = magnesium nitrate

Draw up a table to show the chemical formula and name of each of the following chemical compounds:

K_2CO_3	$FeCO_3$	$NaNO_3$
$Zn(NO_3)_2$	$KHSO_4$	$PbSO_4$
$CaCO_3$	$CuSO_3$	$Ba(NO_3)_2$
$Zn(HSO_4)_2$	$Al_2(SO_3)_3$	$Pb(HCO_3)_2$
$Cu(OH)_2$	$Mg(OH)_2$	$NaNO_3$
$BaSO_4$	$CaSO_4$	$CuCO_3$

SET 3: nonmetal-nonmetal compounds

Rules: 1. if hydrogen is present, it is named first
2. if no hydrogen, the solid nonmetal is named first
3. if only two elements, name of second is shortened and suffix '-ide' is added to the shortened form
4. if only two elements, prefixes 'mon-', 'di-', 'tri-', 'tetra-' etc. are used on the name of the second to indicate how many atoms of it there are in the formula.
5. if more than two elements, first is named and this is followed by the name of the radical.

examples HCl = hydrogen chloride
CO_2 = carbon dioxide
H_2CO_3 = hydrogen carbonate

Draw up a table to show the chemical formula and name of each of the following chemical compounds:

HF	H_2SO_4	NBr_5	PCl_5
H_2SO_3	NBr_3	NO_2	N_2O_5
HI	NO	N_2O_3	CO_2
HNO_3	OCl_2	PBr_3	SF_6
PCl_3	P_2O_5	SO_3	
P_2O_3	SO_2	SCl_6	
CO	SCl_2	HBr	
CCl_4	H_2O	PBr_5	

Naming Chemical Compounds

SET 1: metal-nonmetal compound; two elements only
Rules: 1. the metallic element is named first 2. the name of the nonmetal is shortened 3. the suffix '-ide' is added to this shortened name examples $NaCl$ = sodium chloride $MgBr_2$ = magnesium bromide

Draw up a table to show the chemical formula and name of each of the following chemical compounds:

KCl	$BaCl_2$	Fe_2O_3	CaS
NaI	$AlCl_3$	ZnI_2	CuF_2
MgS	Li_2O	Pb_3O_4	$MnBr_4$
BaO	FeS	$LiBr$	SnO
AlN	PbO	$AgBr$	ZnO
K_2O	MgF_2	$NiCl_2$	AuF

You will need this information to help you name the chemical compounds in sets 2, 3 and 4 ...

- many chemical compounds contain a chemical radical
- a chemical radical is a group of atoms which tend to stay together when the compound reacts with another compound
- the names of many chemical compounds include the name of the radical they contain
- the names of the most common radicals are shown in the table:

Name of radical	Formula of radical as it appears in chemical compounds
ammonium	NH_4
carbonate	CO_3
hydrogen carbonate	HCO_3
hydrogen sulfate	HSO_4
hydrogen sulfite	HSO_3
hydroxide	OH
nitrate	NO_3
sulfate	SO_4
sulfite	SO_3